

Review of the last structure

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

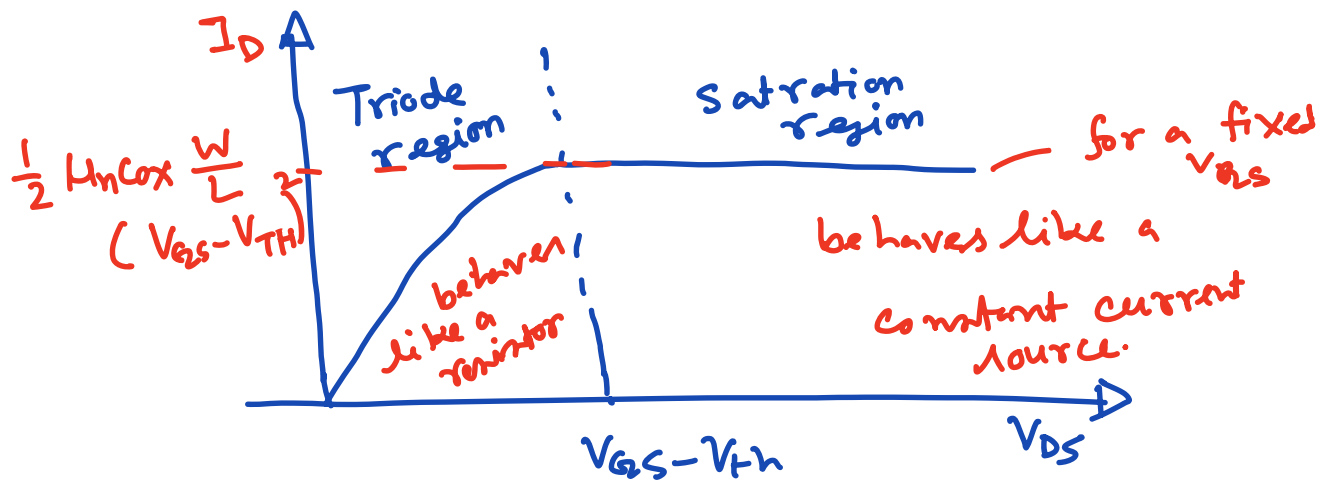
$$V_{DS} \leq V_{GS} - V_{TH}$$

Triode Region

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

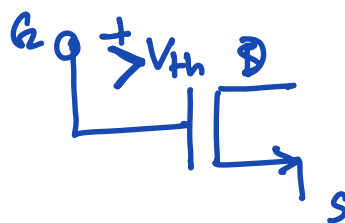
$$V_{DS} > V_{GS} - V_{TH}$$

Saturation region

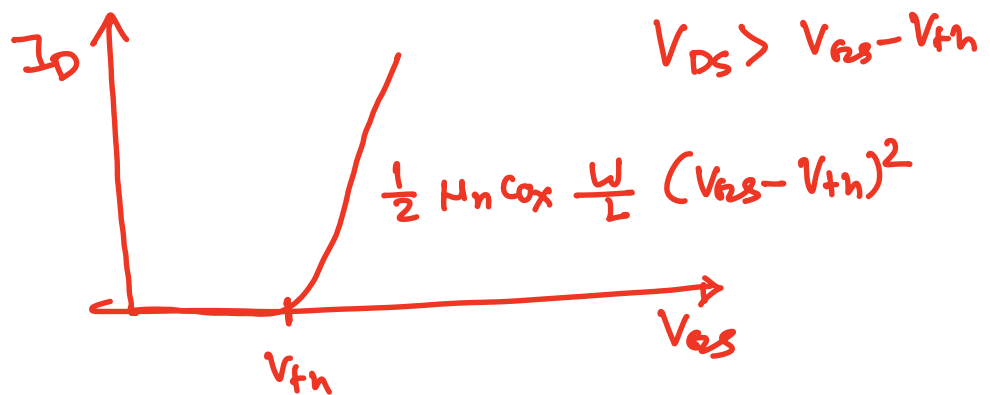
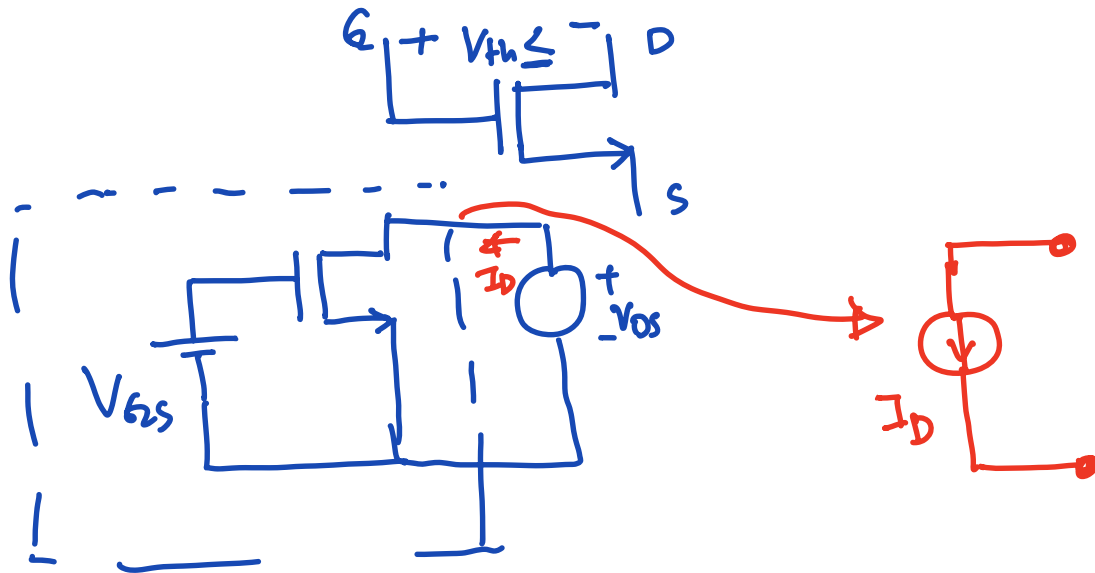


Triode Region

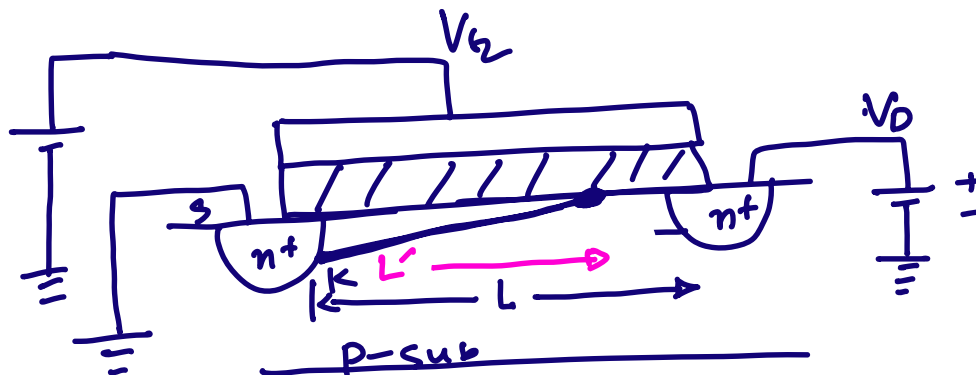
→ When the potential differences between gate and drain is greater than V_{TH} , the MOSFET is in Triode region



→ When the potential difference between gate and drain becomes ~~or~~ equal or less than V_{th} , the MOSFET enters into saturation region (pinch-off)



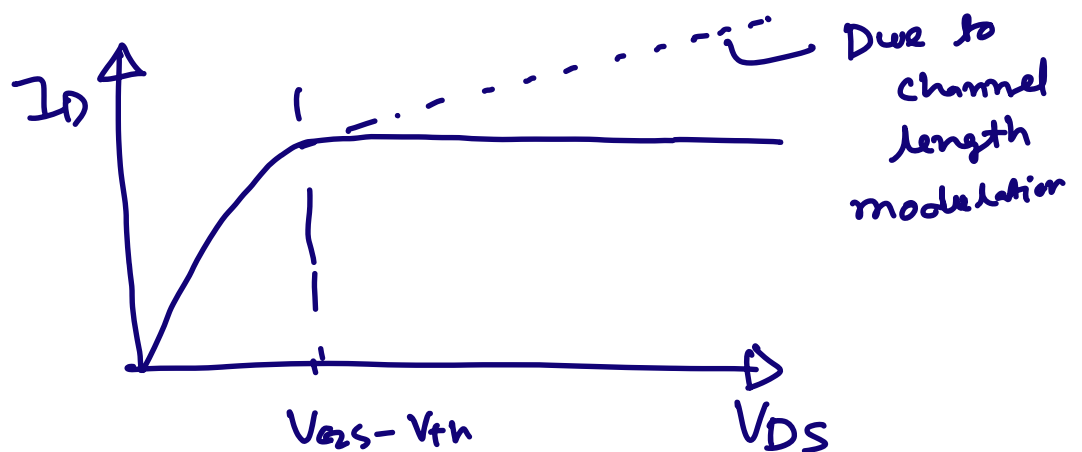
Channel-length Modulation



$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

As V_{DS} increases, $L \downarrow$

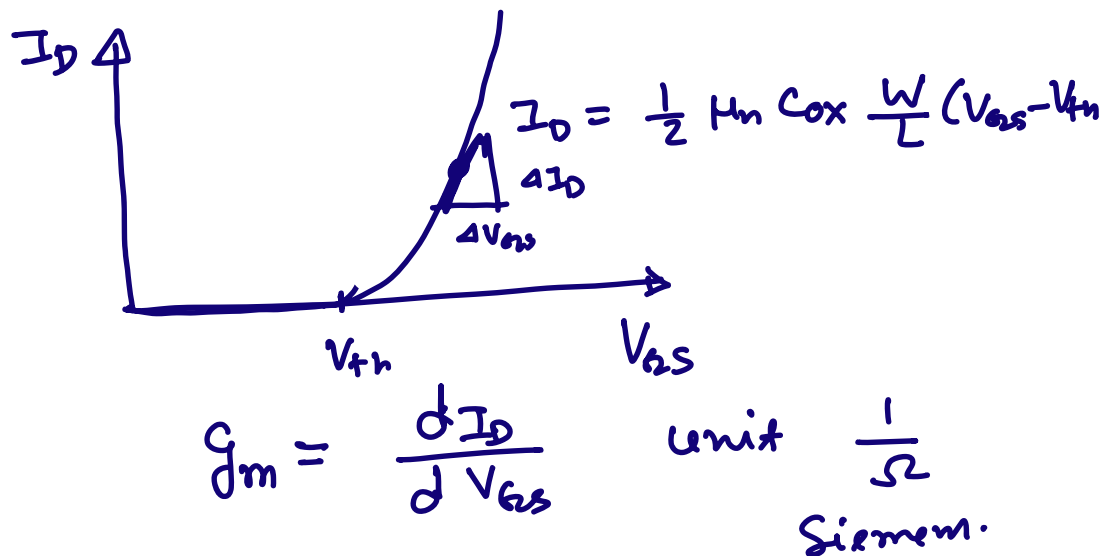
$$\uparrow I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L \downarrow} (V_{GS} - V_{th})^2$$



$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \underbrace{\lambda V_{DS}}_{\text{channel length mod}})$$

λ : channel modulation factor.

MOS Transconductance:



Transconductance is a measure of how strong the drain current changes when the gate voltage changes.

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$g_m = \frac{dI_D}{dV_{GS}}$$

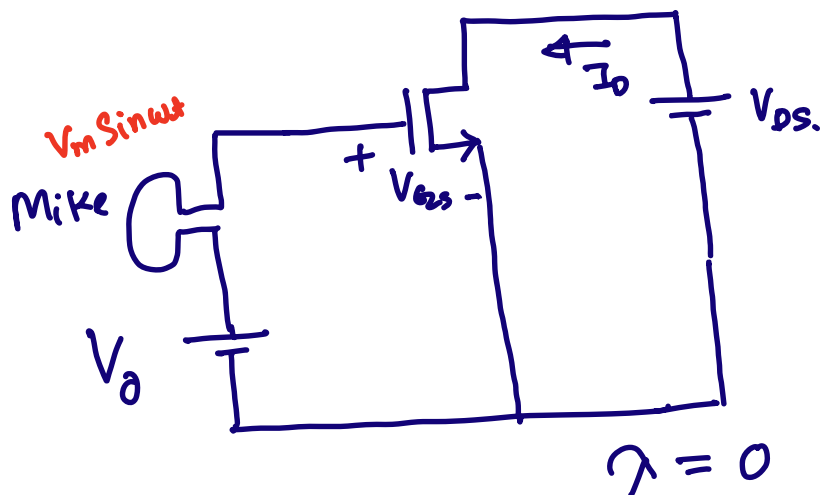
$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})$$

or $g_m = \frac{2 I_D}{V_{GS} - V_{th}}$

Small-signal operation

What is small signal?

What is the small model of MOSFET?



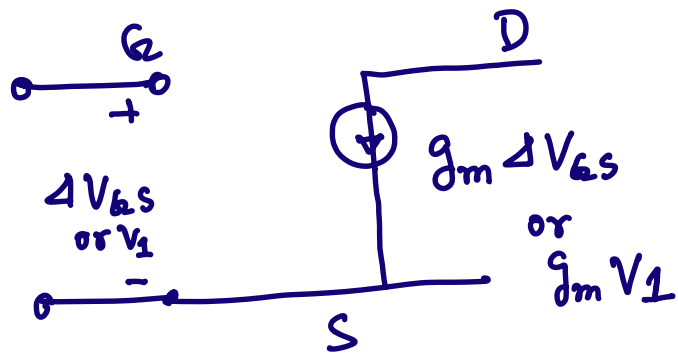
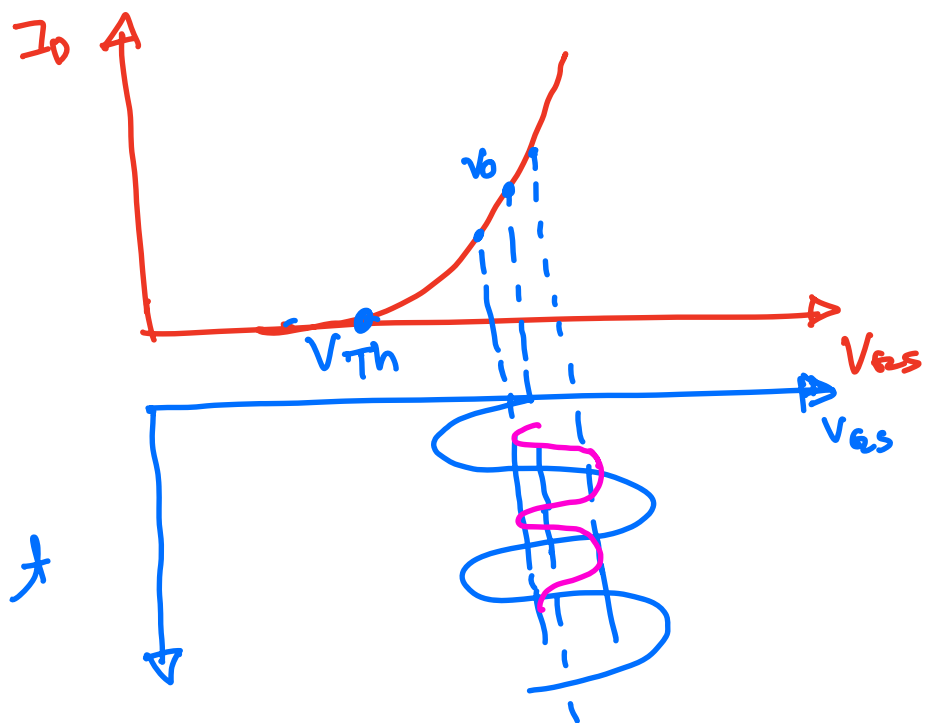
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{gs} - V_{th})^2$$

or $V_0 - V_{th}$ @ bias.

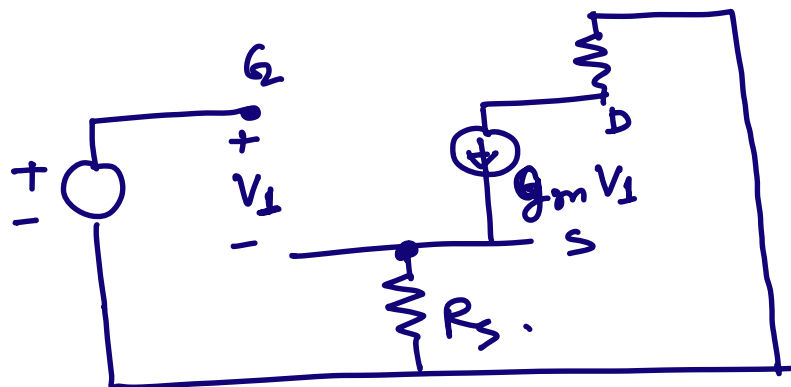
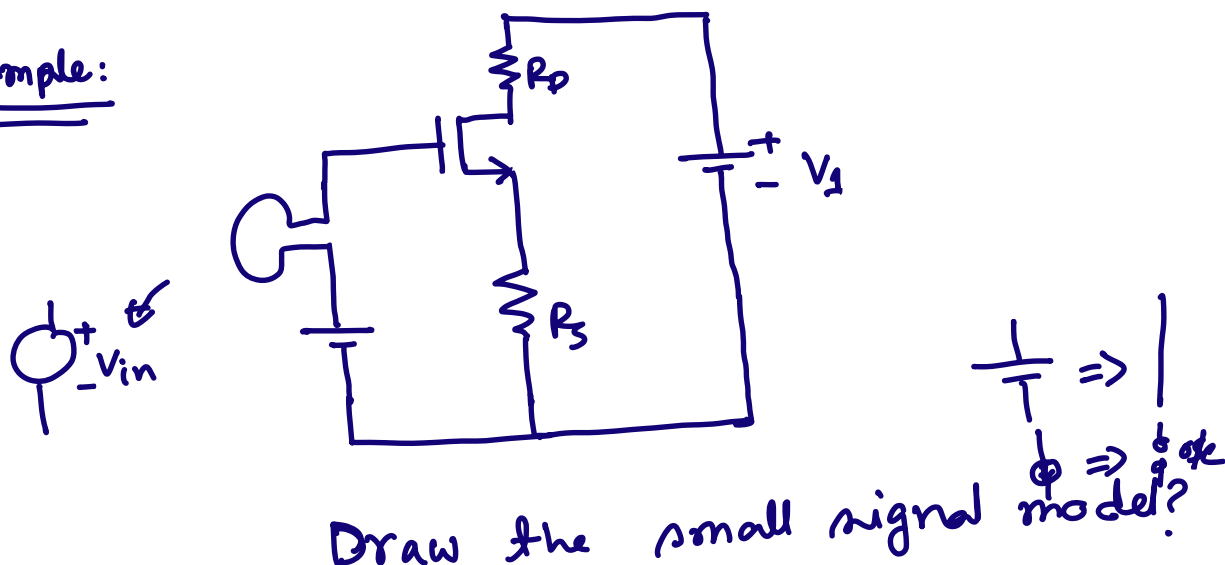
$$= \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (\underbrace{V_0 - V_{th}}_{\text{overdrive potential}} + \underline{V_m \sin \omega t})^2$$

$$\text{Assume } V_m \ll V_0 - V_{th}$$

Small-signal regime.

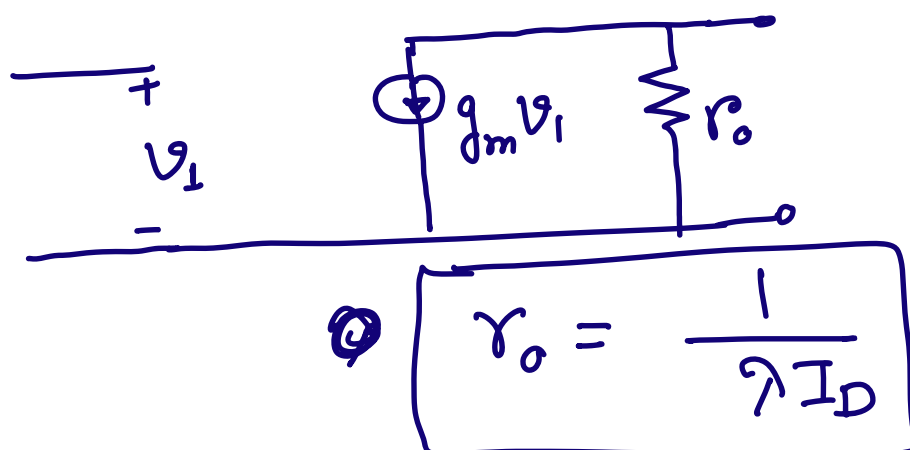


Example:



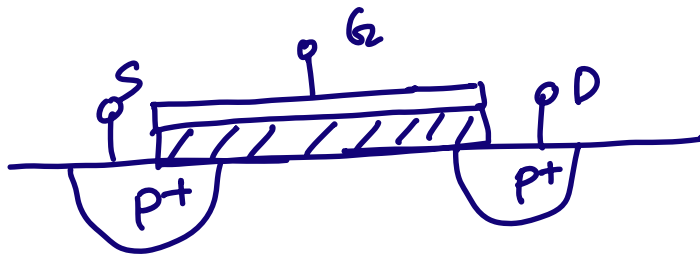
So far we assumed $\lambda = 0$

if $\lambda \neq 0$

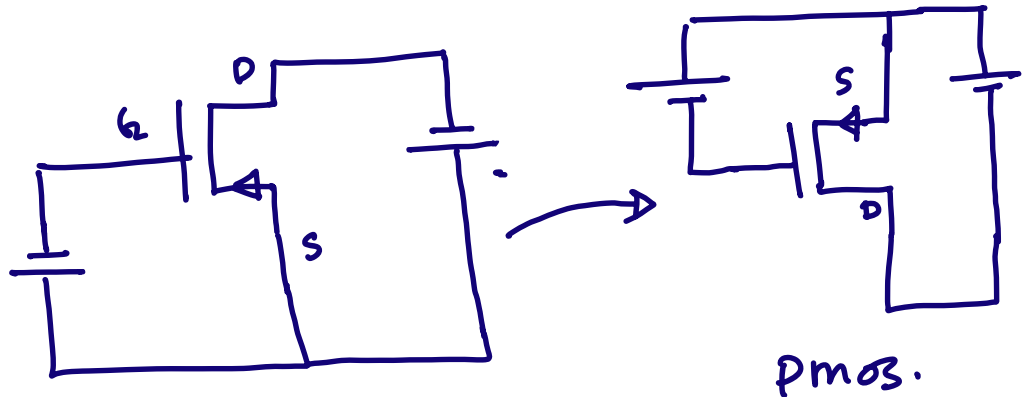


Example 6.14

PMOS Transistor



n-substrate



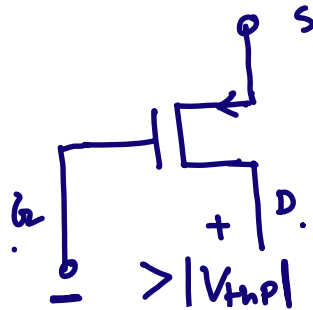
pmos.

In Sat.

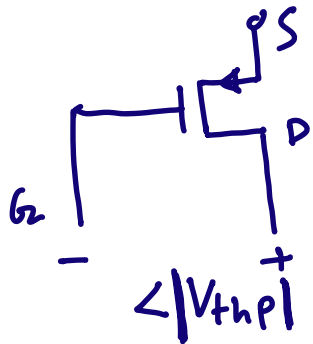
$$I_D = -\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{tn})^2 (1 - \lambda V_{DS})$$

$\lambda > 0$

Regions of operation



Triode.



Saturation

Small-Signal Model:

